

ELEC 477 Operating Systems

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Exam #1

Time: 25 Minutes

Instructions: Total 21 marks. Closed Book.
Answer all questions in the space provided on these pages
If you need extra space, use the extra pages at the end of the exam.
Make sure you indicate to the marker you are using the extra space.

1. (4 marks) Give a brief (one or two line) description of each of the following terms as they were discussed in class.

a. Network Byte Order

The order in which bytes of integer and floating values are transmitted on the network. This is big endian order.

b. UTC

Universal Coordinated Time

c. Failure

When a system cannot provide one or more services. (note the system may not crash but still have a failure)

- also accepted answers about network or hardware failure

d. UDP

User Datagram Protocol - unreliable messaging between internet hosts.

2. (2 marks) Referential Coupling refers to the fact that processes must know (or obtain) the identity of other processes in the system. What are two distributed system architectures discussed in class that do not require referential coupling?

Shared Data Space (aka tuple space)

Publish and Subscribe

- Also accepted ZeroMQ Pipeline arch

3.(3 marks) There are three possible semantics for an RPC call involving how many times the server executes a request. What are they and what do they mean?

At least once - the RPC call is executed by the server one or more times of a request

At most once - the RPC call is executed by the server zero or one times for a request

Exactly once - the RPC call is executed exactly once for a request

4. (4 marks) What are some of the advantages a persistent message oriented middleware has over transient middleware?

Receiver may not be running

Sender and receiver execute independently of another

Once message has been inserted it stays there until received

MOM guaranteed message will be eventually received

Last two are similar

5 (2 marks). What are two advantages of using a Domain Specific Language to marshal and unmarshal parameters when writing the client and server stubs of a remote procedure call?

the most obvious are:

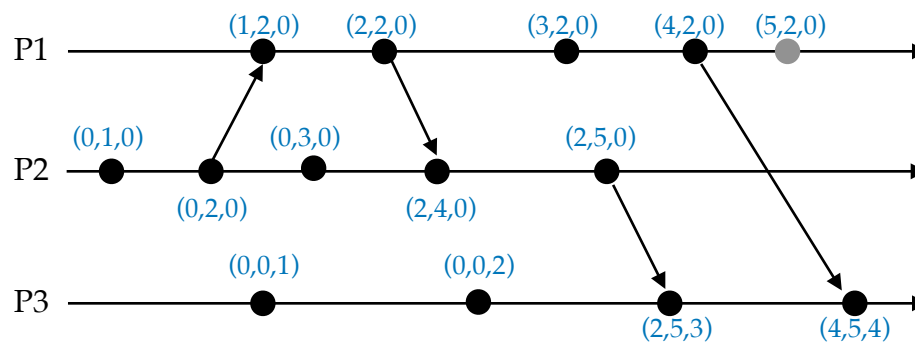
- 1 supports multiple languages from the same specification
2. Automatically adjusts representations between different architectures

Also:

Streamlines developer time as not have to write the code to marshal/unmarshal
Easier to understand than doing it manually

TAs: If they say anything else check with me, there are other advantages

6. (6 marks) The following diagram represents three processes with internal events and messages. Each black circle represents an event.



- a. (4 marks) Label the diagram using a vector clock expressed as a triple (three element tuple).
- b. (2 marks). List **all** of the events (using their vector clock tuples) that are in the casual history of the grey circle.
(0,1,0), (0,2,0), (1,2,0) (2,2,0), (3,2,0), (4,2,0) -> (5,2,0)

Extra Space.

Extra Space.